

Complicações Pós-Operatórias em cirurgias abdominais Específicas



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Complicações – Cirurgias do cólon e delgado

- Anastomose 1ª ??
- Colostomia x Ileostomia
- Início de dieta x Altura ressecção



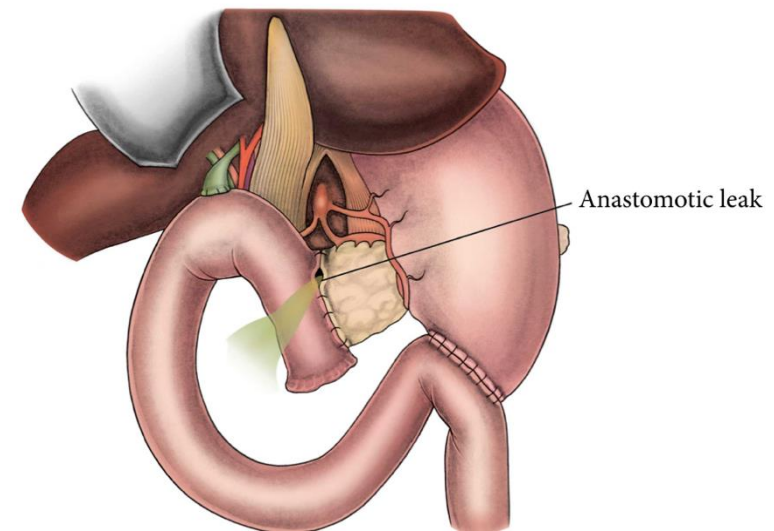
<https://www.endoscopyandsurgery.co.uk/conditions/bowel/small-bowel/>



Deiscência de anastomose (Vísceras ocas)

Precoce x Tardia

- Esofagectomias 5,7-14,3%
- Colocolônica/Ileocólon/ Entero-entero – 0-4%
- + Baixas > Coloanal, Retoanal, Colorectal baixa > 1-20%



<https://doi.org/10.1155/2016/5392923>

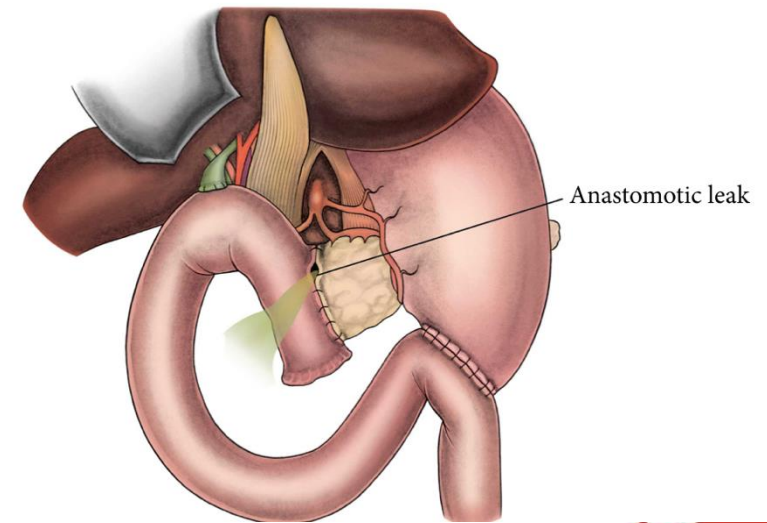


Deiscência de anastomose Intestinal

Febre ? Disfunção Orgânica? Dor abdominal ?

Tomografia com contraste ? VO ?

Radiointervenção x Endoscopia x Cirurgia



<https://doi.org/10.1155/2016/5392923>



Endoscopia

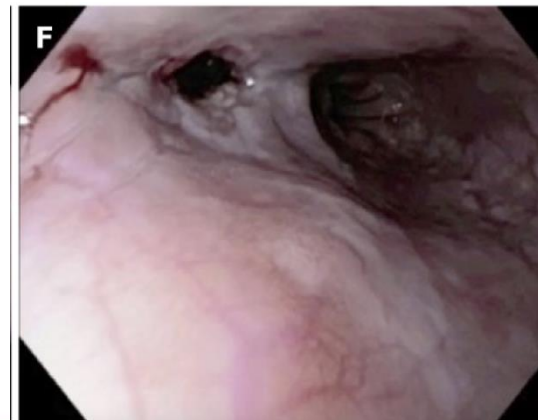
DOI: 10.4253/wjge.v11.i5.329

ISSN 1948-5190 (online)

REVIEW

Role of endoscopic vacuum therapy in the management of gastrointestinal transmural defects

Diogo Turiani Hourneaux de Moura, Bruna Furia Buzetti Hourneaux de Moura, Michael A Manfredi, Kelly E Hathorn, Ahmad N Bazarbashi, Igor Braga Ribeiro, Eduardo Guimarães Hourneaux de Moura, Christopher C Thompson

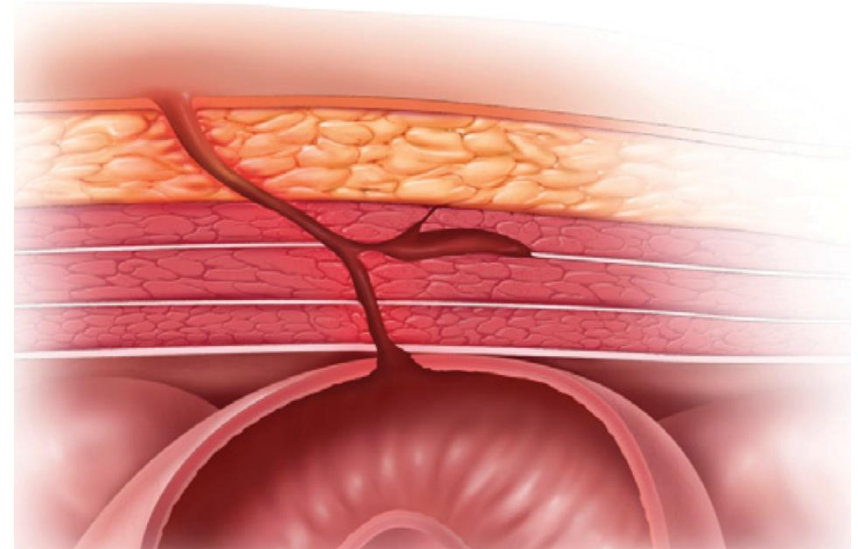


Fístulas abdominais

Víscera > Víscera // Víscera >> Pele // Víscera >> Atmosfera

Baixo débito (<200ml/dia) // médio // Alto (>500ml/dia)

Radiointervenção x Cirurgia



<https://memphisvascular.com/wp-content/uploads/2016/05/3610-SUR-BM-EFPPG-EN-201204m3.pdf>



Fístulas abdominais

Reduzir **Débito**

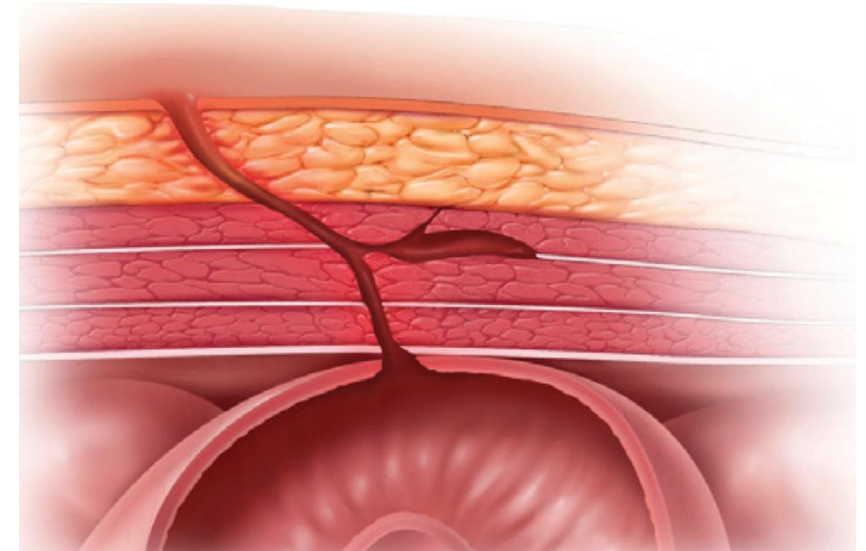
Nutrição (Dieta pós x NPP)

Corrigir **eletrólitos**/água (Alto x intestino médio)

Resolver a **infecção** (ATB + Drenar?)

Proteger a **pele** (Barreira x vácuo)

Fechamento Espontâneo ? 30-40% >> Cirurgia x Timming



<https://memphisvascular.com/wp-content/uploads/2016/05/3610-SUR-BM-EFPPG-EN-201204m3.pdf>



Somatostatina ?

Systematic review and meta-analysis of the role of somatostatin and its analogues in the treatment of enterocutaneous fistula

Philip Stevens, Rhiannon E. Foulkes, Jake S. Hartford-Beynon and Raymond J. Delicata

Conclusion Somatostatin and octreotide improved fistula closure time but only somatostatin improved spontaneous closure rate. *Eur J Gastroenterol Hepatol* 23:912–922 ©

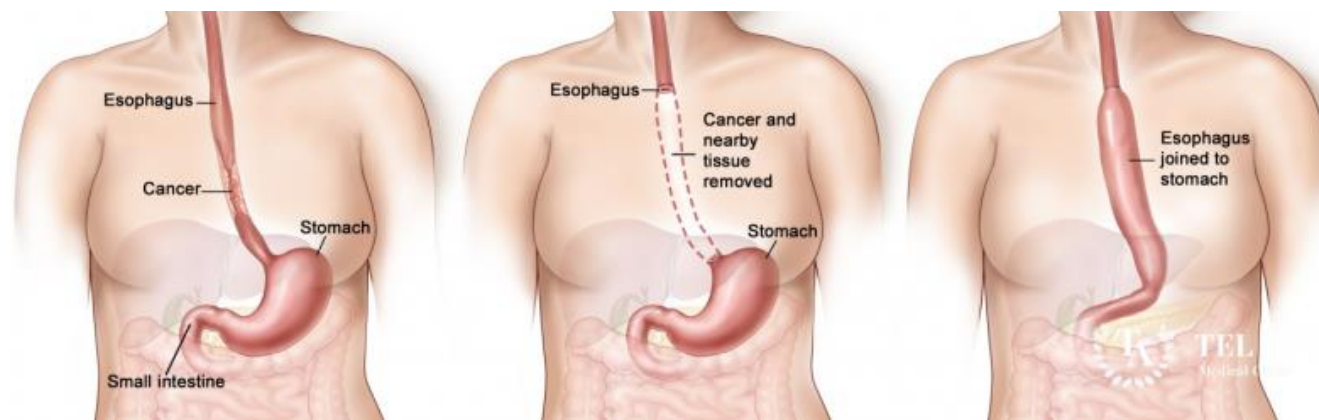
Como Reduzir Débito ? (Além de “jejum”)

- Somatostatina (meia vida 1 minuto!)
 - 6000 Ui / dia
- Octreotide
 - Reduz fluxo 40-93% em 48h
 - 100mcg 8/8h (PMID 15152279)

Ambos reduziram tempo de fechamento! Somatostatina melhorou a taxa de fechamento!
Estudos pequenos ! Alto risco de viés !



Complicações - Esofagectomias



Etiologia x Desnutrição

2 campos x 3 campos

Estômago x Cólon (altura)

Deiscência > 5-15%. >> Febre? FA? Arritmias ? Drenos ?

Complicações Respiratórias x IOP x Dor

Cuidado na Extubação !



Complicações - Hepatectomias

Cirrótico x Não Cirrótico

Localização

Pringle

Sangramento

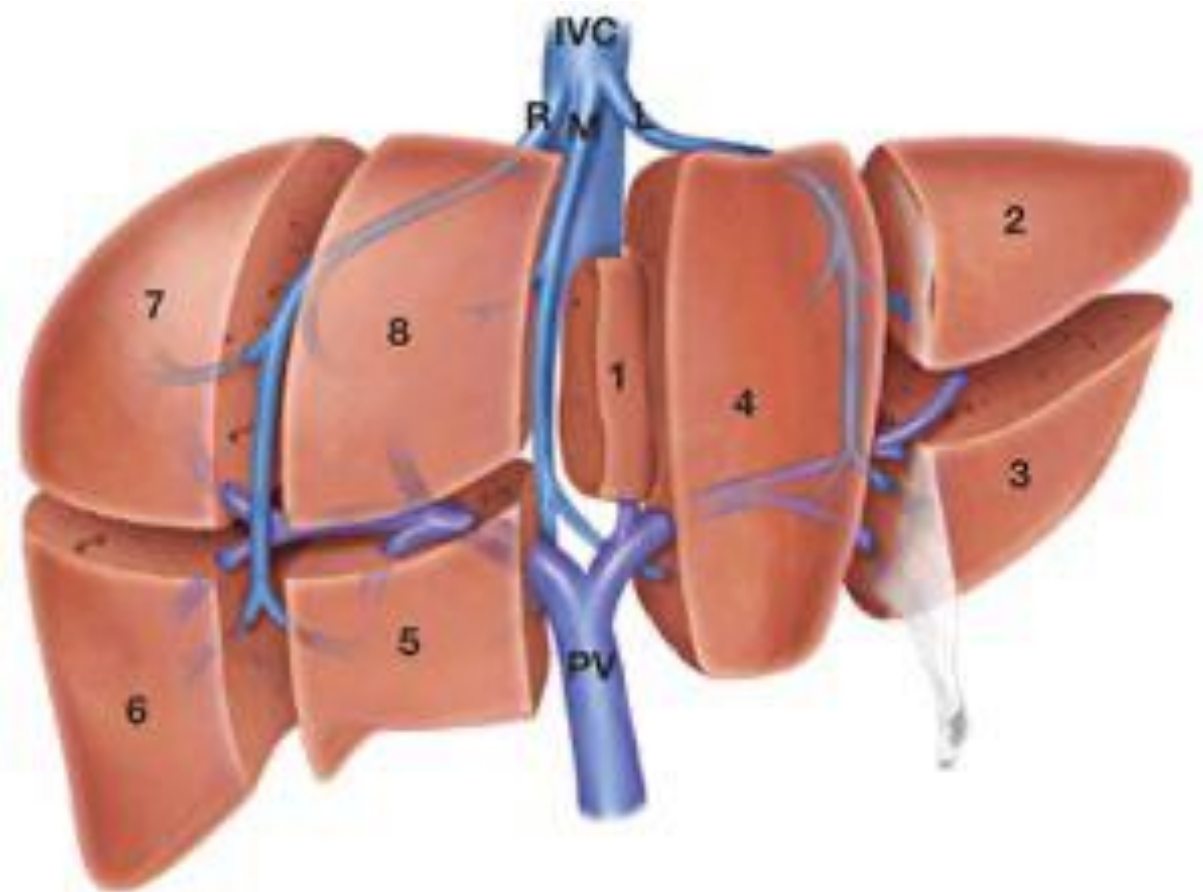


FIG 1 • Couinaud segments with vascular anatomy.

Quem é o Paciente ??

CHC

Outros
(Adenomas, Hemangiomas e etc.)

Cirrose ? + Etiologia

Jovem sem comorbidades ?



Impact of cirrhosis in patients undergoing laparoscopic liver resection in a nationwide multicentre survey

C. Hobeika¹, D. Fuks², F. Cauchy³, C. Goumard¹, O. Soubrane³, B. Gayet², E. Salame⁴, D. Cherqui⁵, E. Vibert⁵ and O. Scatton¹, on behalf of the AFC-LLR-2018 study group*

Table 3 Comparison of intraoperative data and outcomes between patients with and without cirrhosis after propensity score matching

	No cirrhosis (n = 282)	Cirrhosis (n = 282)	P†
Intraoperative data			
Conversion	30 (10.6)	30 (10.6)	0.999
Blood loss (ml)*	376(796)	343(412)	0.860‡
Blood loss ≥ 500 ml	43 (19.0)	39 (18.5)	0.884
Massive blood loss ≥ 1000 ml	20 (9.9)	16 (9.5)	0.888
Duration of operation (min)*	204(99)	185(92)	0.055‡
Inflow clamping	91 (32.3)	109 (39.2)	0.087
Transfusion	15 (6.6)	14 (6.4)	0.936
Outcomes after LLR			
90-day mortality	2 (0.7)	10 (3.5)	0.036§
90-day complications	136 (48.2)	123 (43.8)	0.289
90-day severe complications	18 (6.4)	30 (10.6)	0.096
Cardiorespiratory-related severe complications	3 (1.1)	6 (2.1)	0.338§
Liver-related severe complications	11 (3.9)	12 (4.3)	0.825
Bleeding	1 (0.4)	3 (1.1)	0.373§
Bile leak	6 (2.1)	2 (0.7)	0.286§
ISGLS grade B/C PHLF	1 (0.4)	7 (2.5)	0.068§
Postoperative ascites	5 (1.8)	22 (7.8)	0.001
Reoperation	4 (1.4)	7 (2.5)	0.352
ICU for ≥ 1 day	84 (32.4)	105 (43.4)	0.011
Duration of hospital stay (days)*	8(5)	11(32)	0.018‡
Prolonged hospital stay (≥ 9 days)	67 (23.8)	84 (30.1)	0.090
Readmission	27 (9.6)	25 (8.9)	0.771

IOP Semelhante

PO com + Mortalidade
+ Insuficiência
+ UTI

Cirrose



CIRROSE

Original article

Effects of intermittent Pringle's manoeuvre on cirrhotic compared with normal liver

Y. Sugiyama, Y. Ishizaki, H. Imamura, H. Sugo, J. Yoshimoto and S. Kawasaki

Department of Hepatobiliary–Pancreatic Surgery, Juntendo University School of Medicine, 2-1-1 Hongo, Bunkyo-ku, Tokyo 113-8421, Japan

Correspondence to: Dr Y. Ishizaki (e-mail: ishizaki@juntendo.ac.jp)

Table 4 Assessment of postoperative liver function

	Normal liver (<i>n</i> = 172)	Chronic hepatitis (<i>n</i> = 59)	Liver cirrhosis (<i>n</i> = 97)	<i>P</i> [*]
Maximum total bilirubin level (mg/dl)	1.25(0.58)	1.47(0.48)	1.52(0.55) [†]	< 0.001
Minimum prothrombin time (%)	74(13)	71(13)	68(10) [†]	< 0.001
Minimum albumin level (g/dl)	4.0(0.4)	4.1(0.4)	3.8(0.5) ^{†‡}	< 0.001
Minimum cholinesterase level (units/l)	558(239)	509(249)	357(179) [†]	< 0.001

+ disfunção PO !

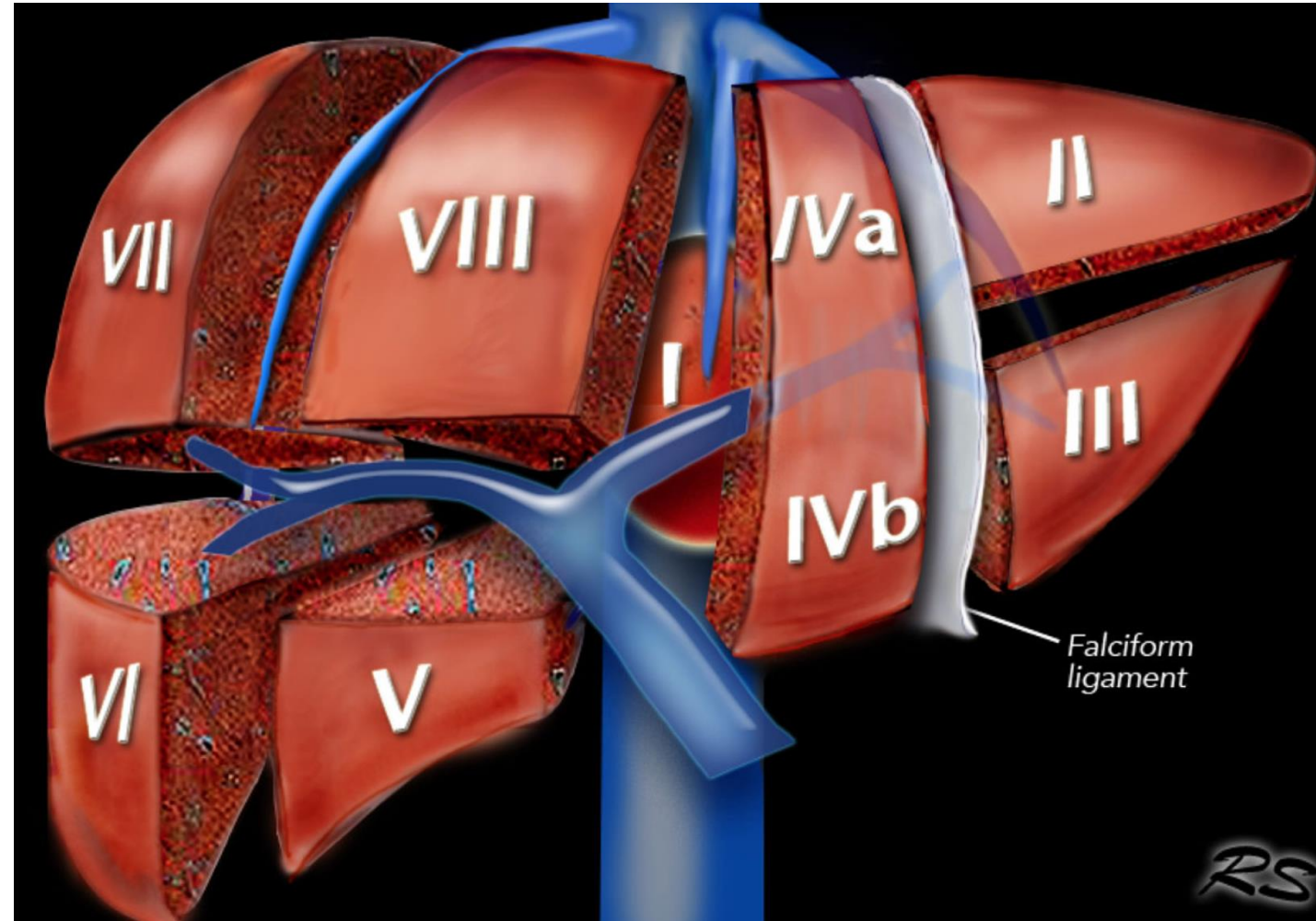
Values are mean(s.d.). ^{*}One-way ANOVA. [†]*P* < 0.050 *versus* normal liver, [‡]*P* < 0.050 *versus* chronic hepatitis (Tukey's test).

^{*}*P* = 0.021, [†]*P* = 0.021, [‡]*P* < 0.001 (test of *t* value)



Tumores Hepáticos

Localização



<https://radiologyassistant.nl/abdomen/liver/segmental-anatomy>



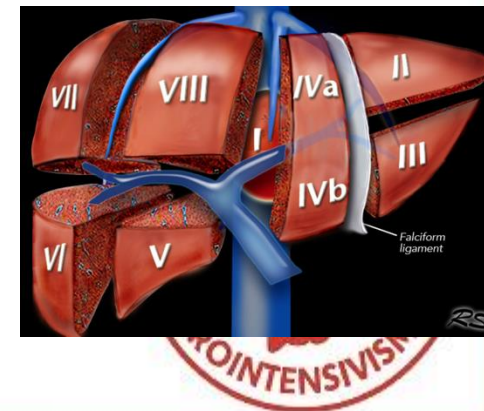
Risk Stratification of 7,732 Hepatectomy Cases in 2011 from the National Clinical Database for Japan

Akira Kenjo, MD, PhD, Hiroaki Miyata, PhD, Mitsukazu Gotoh, MD, PhD, Yukou Kitagawa, MD, PhD, FACS, Mitsuo Shimada, MD, PhD, FACS, Hideo Baba, MD, PhD, FACS, Naohiro Tomita, MD, PhD, Wataru Kimura, MD, PhD, Kenichi Sugihara, MD, PhD, Masaki Mori, MD, PhD, FACS

Table 2. Surgical Procedures for Hepatectomy

Hepatectomy	Involved segments	Cases, n	30-d Mortality			90-d In-hospital mortality		
			Deaths, n	%	p Value	Deaths, n	%	p Value
Two segments		4007	74	1.8	0.26	157	3.9	0.73
Left	S2,3,4	797	8	1.0	0.033	11	1.4	<0.00
Right	S5,6,7,8	1359	37	2.7	0.06	75	5.5	0.14
Central	S4,5,8	209	1	0.5	0.13	6	2.9	0.59

Menor mortalidade > Hepatectomia E



Quanto vai sobrar ? 30%?

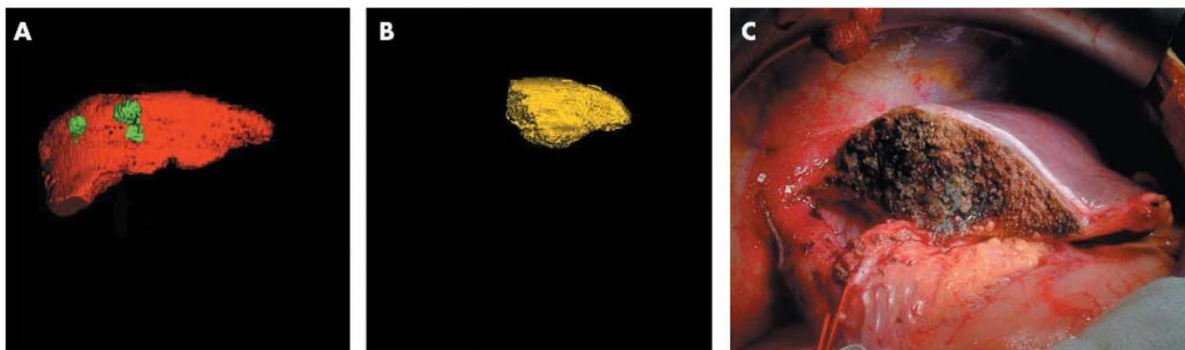


Figure 1 (A–C) Three dimensional liver volumetry of an extended right hepatectomy with caudate excision. (A) Total liver volume (TLV = 1133 ml; red colour); tumour volume (TuV = 34 ml; green colour); and total functional liver volume (TFLV = 1099 ml). (B) Residual liver volume (RLV = 293 ml yellow colour) from three dimensional liver model after virtual resection; relative residual liver volume (%RLV = 26.7%). (C) Intraoperative view of the residual liver following resection.

LIVER

The value of residual liver volume as a predictor of hepatic dysfunction and infection after major liver resection

M J Schindl, D N Redhead, K C H Fearon, O J Garden, S J Wigmore, on behalf of the Edinburgh Liver Surgery and Transplantation Experimental Research Group (eLISTER)

Gut 2005;54:289–296. doi: 10.1136/gut.2004.046524

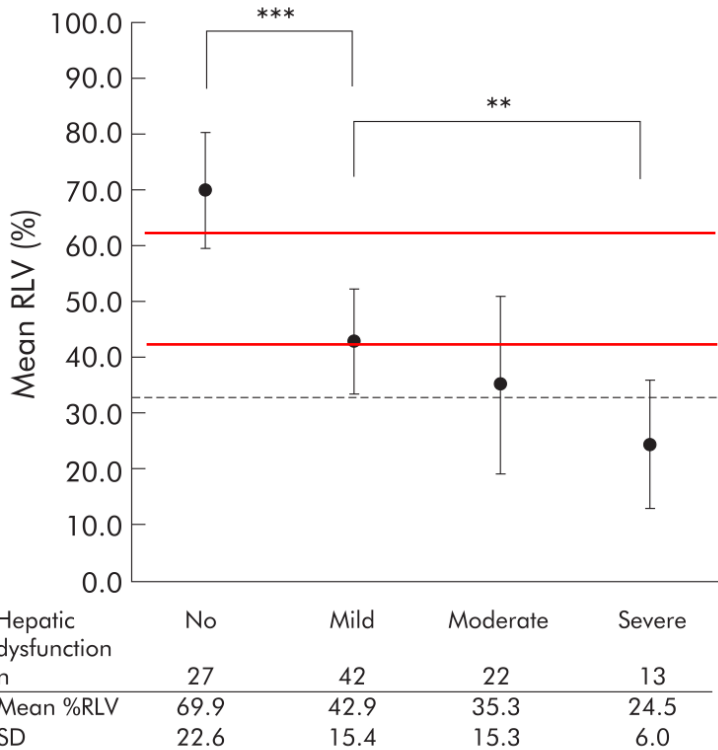
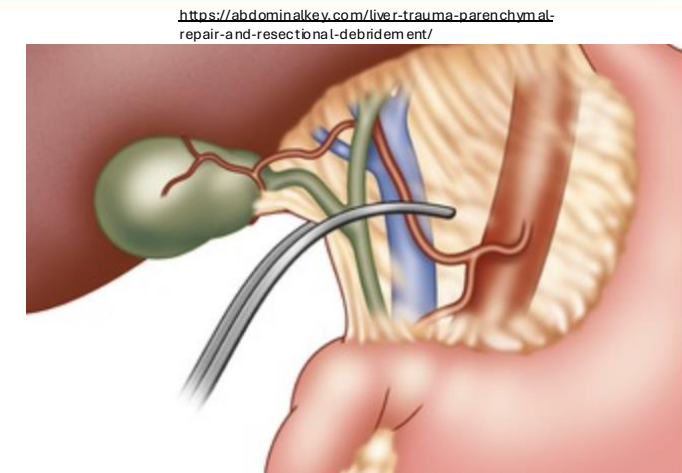


Figure 3 Mean (SD) relative residual liver volume (%RLV) in patients with no, mild, moderate, and severe hepatic dysfunction following liver resection (one way between group ANOVA; **p = 0.005, ***p < 0.0001). Reference line indicates 33% RLV.

Manobra de Pringle



< sangramento
> Isquemia

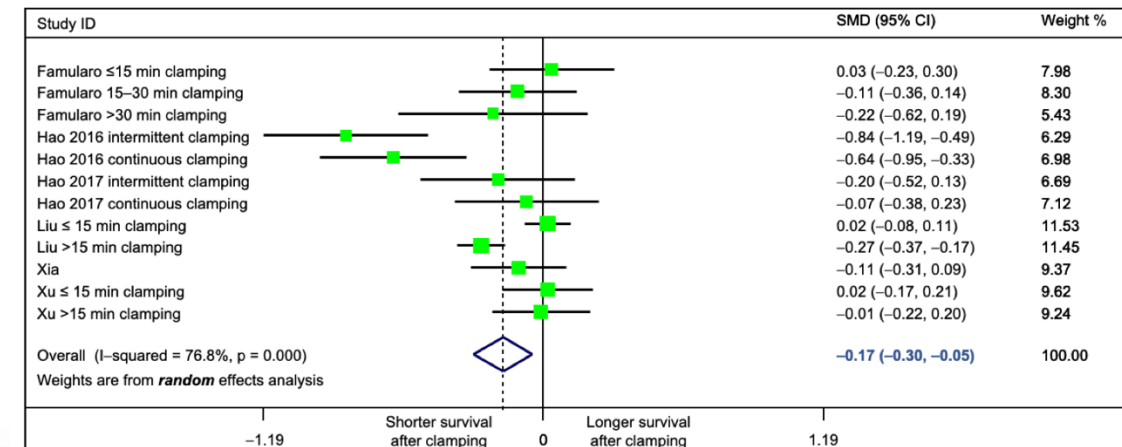
MARCADOR DE COMPLEXIDADE?



Article

Shorter Survival after Liver Pedicle Clamping in Patients Undergoing Liver Resection for Hepatocellular Carcinoma Revealed by a Systematic Review and Meta-Analysis

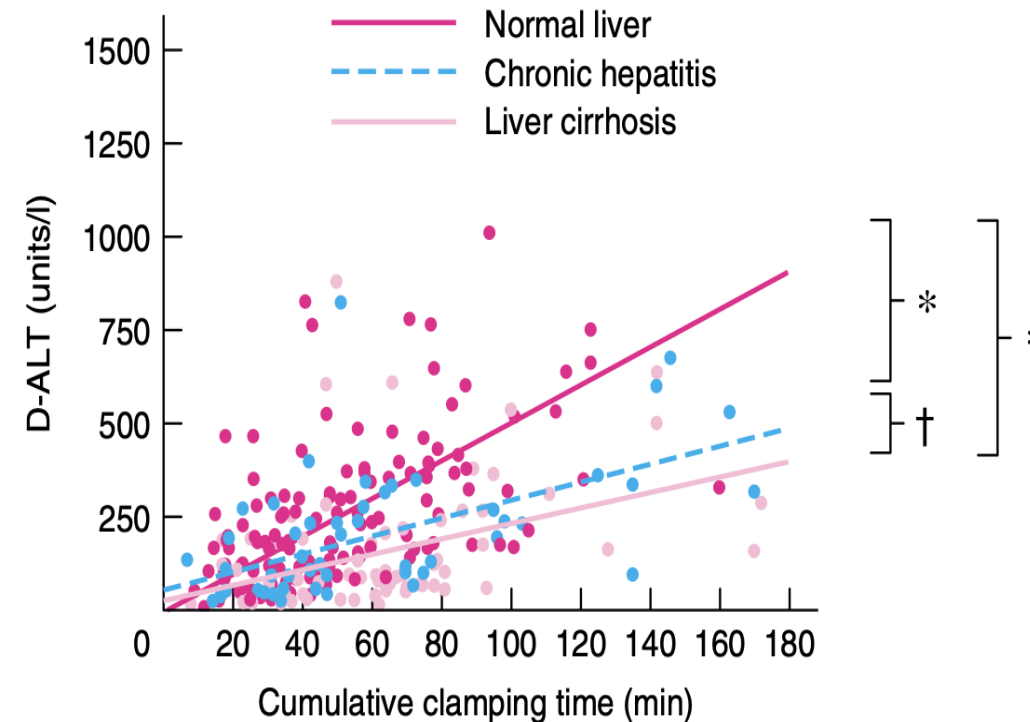
Charles-Henri Wassmer *, Beat Moeckli *, Thierrv Bernev, Christian Toso and Lorenzo A. Orci



Effects of intermittent Pringle's manoeuvre on cirrhotic compared with normal liver

Y. Sugiyama, Y. Ishizaki, H. Imamura, H. Sugo, J. Yoshimoto and S. Kawasaki

Department of Hepatobiliary–Pancreatic Surgery, Juntendo University School of Medicine, 2-1-1 Hongo, Bunkyo-ku, Tokyo 113-8421, Japan
Correspondence to: Dr Y. Ishizaki (e-mail: ishizaki@juntendo.ac.jp)



> tempo Pringle
> Isquemia

Sangramento

Table 4. Prevalence of Morbidity with Hepatectomy Outcomes

	Entire study population (n = 7,732)		Hepatectomy outcomes groups			
	n	%	30-d Mortality (n = 157)		90-d In-hospital mortality (n = 309)	
Postoperative outcomes			n	%	n	%
30-day mortality	157	2.0				
Operative mortality	309	4.0				
Readmission within 30 d	138	1.8	2	1.3	5	1.6
Reoperation within 30 d	203	2.6	30	19.1	53	17.2
Surgical complication						
Anastomotic leak	137	1.8	6	3.8	32	10.4
Bile leak	620	8.0	19	12.1	69	22.3
Wound dehiscence	90	1.2	8	5.1	24	7.8
Transfusion >5 U	327	4.2	77	49.0	138	44.7

Alta Mortalidade !



Impact of the Prolonged Intermittent Pringle Maneuver on Post-Hepatectomy Liver Failure: Comparison of Open and Laparoscopic Approaches

Shunsuke Doi¹ · Satoshi Yasuda¹  · Daisuke Hokuto¹ · Naoki Kamitani¹ · Yasuko Matsuo¹ · Takeshi Sakata¹ · Satoshi Nishiwada¹ · Minako Nagai¹ · Kota Nakamura¹ · Taichi Teraï¹ · Yuichiro Kohara¹ · Masayuki Sho¹

> Pringle
> Disfunção Hepática PO

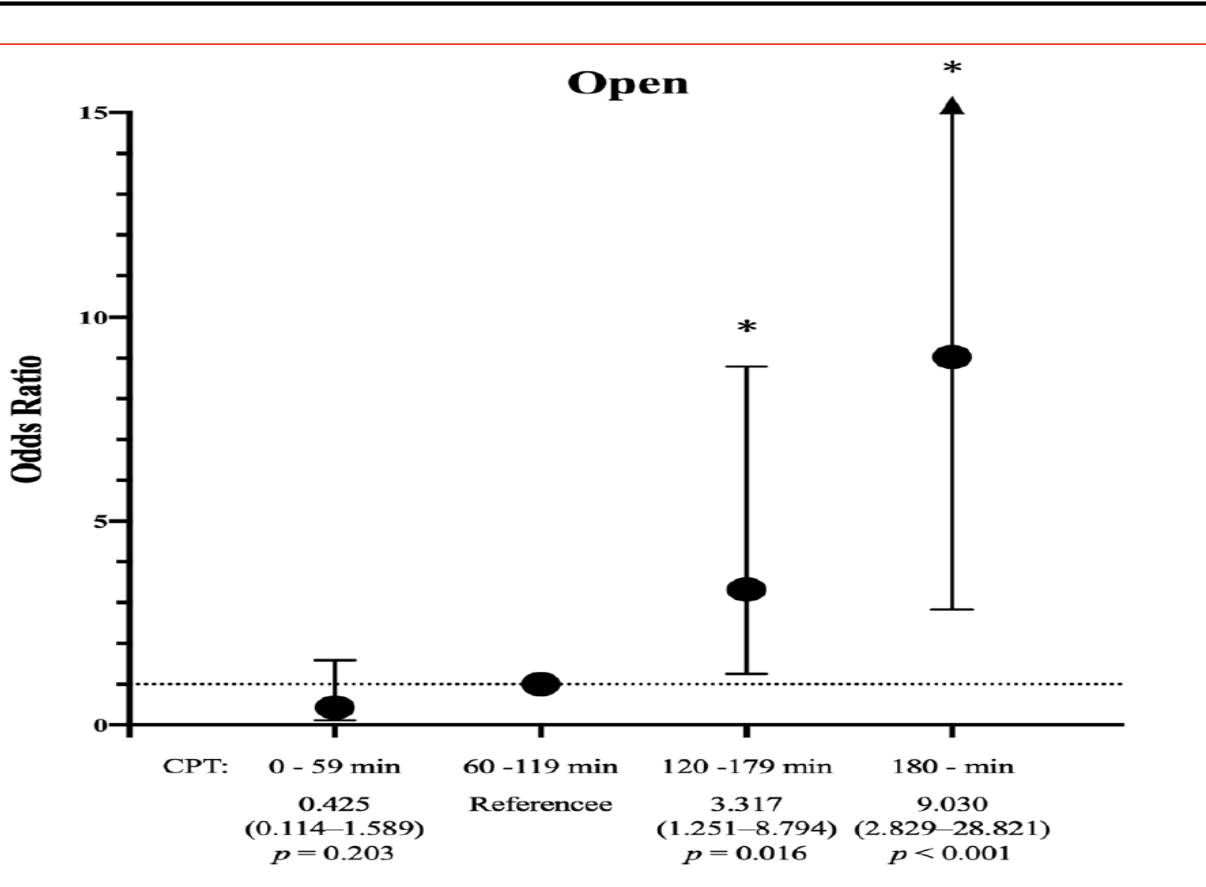


Fig. 2 Odds ratios for PHLF according to CPT

Fístula Biliar

HepatoBiliary Surgery and Nutrition, Vol 9, No 2 April 2020

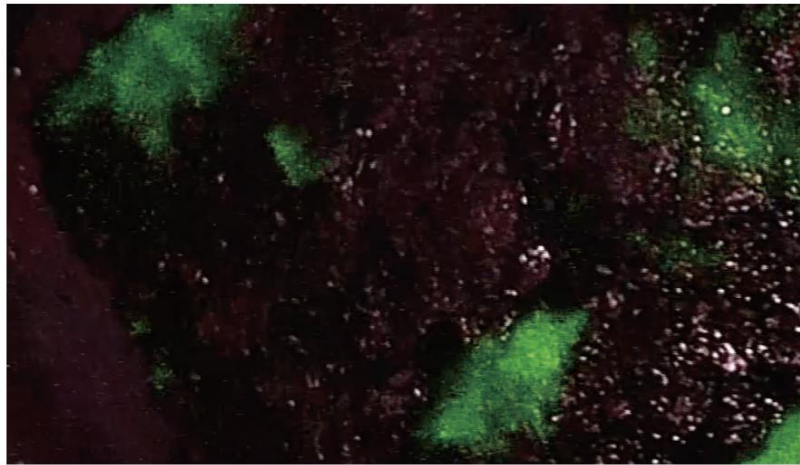


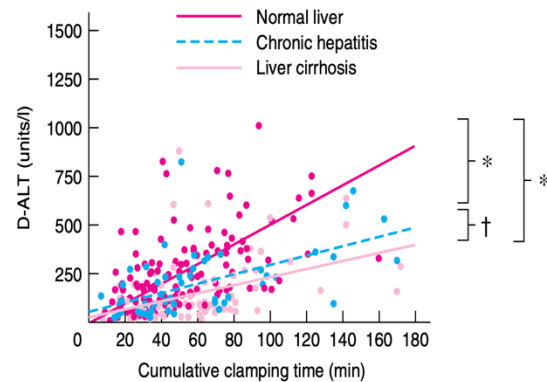
Figure 1 Bile ducts visualization on the cut surface of the liver on ICG fluorescent imaging. ICG, indocyanine green.

Porejamento da superfície Cruenta ??

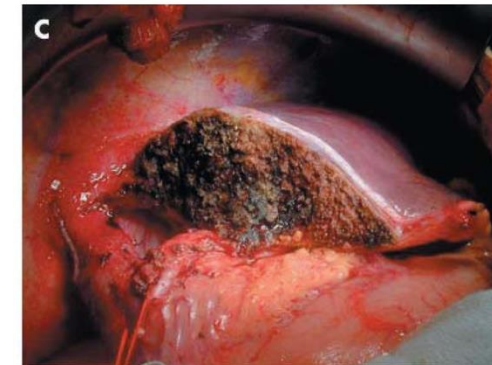
Anastomose ??



Insuficiência Hepática Aguda PO



Isquemia sobre o que sobrou



Sobrou Pouco

Qual Vasopressor ?



Complicações - Pancreatectomias

Fístula Pancreática 10-15% ?

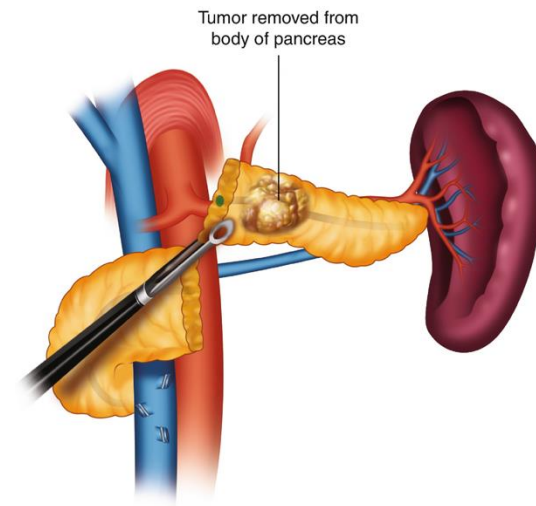
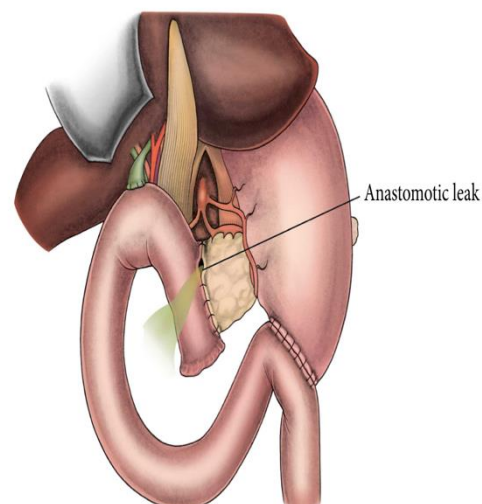
Pancreatite

Sangramentos (8-10% das GDPs). (30-50% mortalidade)

- Investigar se sangra após 24-48h !
- Mesmo se pequeno ! (sangramento sentinela em 30%)
- Pseudo aneurisma (2.5%)

GDP > mortalidade 90d de 5% // Morbidade 30-50%

https://link.springer.com/chapter/10.1007/978-3-030-41350-7_26



Fístula Pancreática

30-50% das GDPs, 70% das mortes !

Diagnóstico:

- Amilase dreno > 3x sérico no D3

Investigar precoce (Sirs + Suspeita > TC) + Manejo menos invasivo

TTO conservador resolve 90-95% (5-6 semanas)

- Dieta, drenos, Somatostatina ?
- Dieta VO se assintomático e baixo débito (vigiar)

THE LANCET

Submit A

ARTICLES | VOLUME 399, ISSUE 10338, P1867-1875, MAY 14, 2022

Algorithm-based care versus usual care for the early recognition and management of complications after pancreatic resection in the Netherlands: an open-label, nationwide, stepped-wedge cluster-randomised trial

F Jasmijn Smits, MD * • Anne Claire Henry, MD * • Prof Marc G Besselink, PhD • Prof Olivier R Busch, PhD • Prof Casper H van Eijck, PhD • Mark Arntz, PhD • et al. [Show all authors](#) • [Show footnotes](#)

Published: April 28, 2022 • DOI: [https://doi.org/10.1016/S0140-6736\(22\)00182-9](https://doi.org/10.1016/S0140-6736(22)00182-9) • [Check for updates](#)



Complicações – Bariátricas

Comorbidades

Eficácia

Deiscência x Peritonite ?

TEP

Hipoxemia / Hemorragia / IAM

Bypass gástrico
(gastroplastia em "Y de Roux")



Gastrectomia Vertical



Banda Gástrica Ajustável



Duodenal Switch



<https://psicologiacaatalao.com.br/2017/03/29/cirurgia-bariatrica-riscos/>



Complicações – Bariátricas

High Mortality Rate for Patients Requiring Intensive Care After Surgical Revision Following Bariatric Surgery

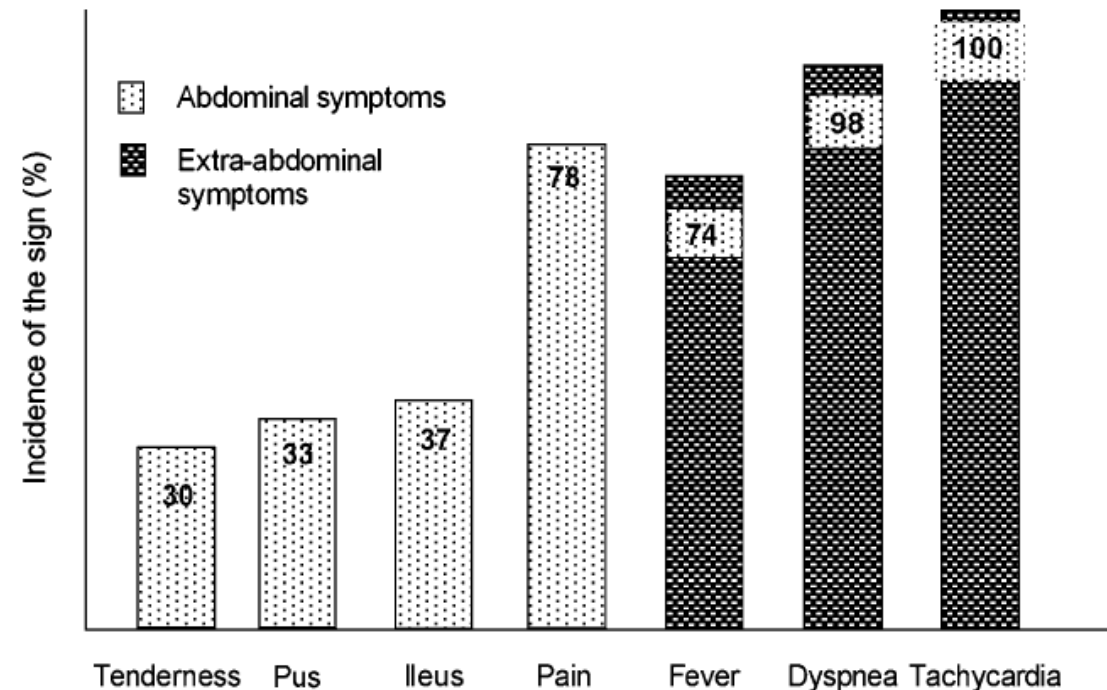


Fig. 1 Physical signs recorded during the postoperative course after bariatric surgery





NIH Public Access

Author Manuscript

Obes Surg. Author manuscript; available in PMC 2014 February 14.

Published in final edited form as:

Obes Surg. 2013 November ; 23(11): 1874–1879. doi:10.1007/s11695-013-1073-1.

NIH-PA Author Man

Incidence of Venous Thromboembolism After Bariatric Surgery: A Population-Based Cohort Study

David A. Froehling, MD, Paul R. Daniels, MD, Karen F. Mauck, MD, Maria L. Collazo-Clavell,

Venous Thromboembolism Events After Bariatric Surgery

Patient	Age, y	Sex	Type of Surgery	Thromboembolic Event	Days after Surgery
1	56	Female	Open RYGP	DVT and PE	20
2	66	Male	Open RYGP	PE	2,044
3	52	Female	Open RYGP	DVT	16
4	57	Female	Open RYGP	DVT	12
5	53	Male	Open RYGP	PE	41
6	41	Male	Open revision	PE	4
7	41	Female	Open revision	PE	15
8	48	Female	Open RYGP	DVT and PE	15
9	44	Female	Laparoscopic RYGP	DVT	17

Abbreviations: DVT, deep vein thrombosis; PE, pulmonary embolism; RYGP, Roux-en-Y gastric bypass.



Incidência de TEP

0.3% em 7 dias

1.9% em 30 dias

2.1% em 90 dias

2.1% em 6 meses

Common postbariatric surgery emergencies for the acute care surgeon: What you need to know

Vincent Cheng, MD, Kamran Samakar, MD, Adrian B. Dobrowolsky, MD, James D. Nguyen, MD, Stuart A. Abel, MD, Andrea Pakula, MD, Andrew Bernard, MD, and Matthew J. Martin, MD, FACS, FASMBS,
Los Angeles, California

GASTRIC BYPASS COMPLICATIONS

Early Complications (≤ 30 Postoperative Days): Leaks, Bleeding, and Early Small Bowel Obstruction

The top three major concerns early after an RYGB should be leak, leak, and leak. The most common leak location is at the gastrojejunal anastomosis, but any anastomosis or staple line can leak (gastric remnant or jejunojejunostomy).^{5,25} Leaks may

